

We claim that the system is consistent since there are only 4 pivot columns and in rref form, all the elements in the column containing the pivot, apart from the pivot itself is 0. The columns containing the free variables are not all zeros. $\therefore$ The augmented matrix can never be of the form $0\ 0\ 0\ 0\ 0\ 0\ b$ where $b \neq 0$ and hence is not inconsistent.

Book's Solution (Fantastic)

Since the coefficient matrix has four pivots, there is a pivot in every row of the coefficient matrix. This means that when the coefficient matrix is row reduced, it will not have a row of zeros, thus the corresponding row reduced augmented matrix can never have a row of the form $[0\ 0\ \dots\ 0\ b]$, where b is a nonzero number. By Theorem 2, the system is consistent. Moreover, since there are seven columns in the coefficient matrix, and only four pivot columns, there will be three free variables resulting in infinitely many solns.

Exercises 1.2

In Exercises 1 and 2, determine which matrices are in reduced echelon form and which others are only in echelon form.

1a.
$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Soln:

For a matrix to be in echelon form, the following conditions must be satisfied:

1. All non-zero rows are above any rows of all zeros. Here there are no zero rows. So, this condition is ~~vacuously~~ satisfied.

2. Each leading entry of a row is in a column to the right of the leading entry of the row above it.

Row 1's leading entry: 1 in Column 1
Row 2's leading entry: 1 in Column 2
Row 3's leading entry: 1 in Column 3

$\therefore$ The condition is satisfied.